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Interim Report on Investigations on Multi Disease Prediction in Dairy Cattle Using Machine Learning on French Dairy Cattle Dataset

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1 Background

Development of agricultural sector is an integral part of every nations economical development which directly impacted GDP. Apart from the farming and corps development, livestock farming forms the inevitable part of agricultural economies in European Union, of which the 1.4% of total GDP was contributed by agriculture [1] . Milk production from dairy cattle accounts to 14% of the total agricultural production after fruits and vegetables [2]. In this regard, the health and well-being of cattle directly impacts not only the the global economic development but also the food security and the animal welfare. This in turn brings out the significance of health care of animals, which can be effectively solved by proper earlier systematic disease diagnosing systems.

The timely disease diagnosis via. conventional methods such as veterinarian intervention has many challenges. One such major challenge is the location of the cattle farms, majority of them are in rural areas where the access to the medical facility and veterinary specialists are limited. The inability of the farmers to detect the disease symptoms due to the lack of knowledge and awareness adds to the other challenge associated with the timely diagnosis of diseases such as mastitis, oestrus, lameness, and calving [3] [4]. This in turn results in the preventable disease conditions progressing to severe sufferings for the cattle and which may result in mortality resulting in huge economic losses.

Therefore this work aims to address these challenges by developing an data driven machine learning algorithm for disease prediction using behavioral and activity-level data collected from cattle. This prediction is carried out by multi-output classification, which allows simultaneous prediction of multiple diseases the same time with respect to the behavioral data of the cattle. The outcome of this approach which is the early prediction of the disease can effectively offer a scalable solution for early warning enabling farmers and veterinarians to intervene promptly and reduce the severity and spread of diseases.

The main focus of this current study is to predict four key diseases—mastitis, oestrus, lameness, and calving using the structured behavioral data collected from the french diary dataset. In order to address the data imbalance and feature correlation, oversampling methods such as SMOTE will be checked. The modeling starts with the traditional models like Random Forest and progresses towards the deep learning architectures such as MLP, aiming to optimize both accuracy and interpretability. The effectiveness of time series approach also will be checked in the current work to analyze the effect of temporal patterns involved using method such as LSTM.

1.1 Objectives of the Study

The main objective of the study is to develop a multiple output disease prediction model using machine learning algorithm from the available merged datasets of cattle from the French agricultural database.

This study aims to:

- Select the ample amount of data and carry out the data cleaning and misisng data handling.
- Carry out EDA for the data to analyze the trends of the data.
- Subject the data to conventional models such as random Forest, Gradient Boosting, Gradient descent, and deep learning methods such as MLP, Tab net.

- Check the possibility of time series analysis via. neural networks such as LSTM to understand the importance of underlying temporal pattern.
- Investigate the effect of oversampling method such as SMOTE which could compensate for the imbalance in the dataset.
- Deployment of the model and artefact building.

1.2 Strengths of This Study

- Larger dataset (300+ cows) with behavioral data with time characteristics
- Multi-output prediction, enabling simultaneous detection of multiple health events at the same time compared toindividual training performed by previous studies.
- Data imbalance handling via. SMOTE (Synthetic Minority Oversampling Technique) integration and their effectiveness checking in the disease prediction cases.
- Comparison between traditional models and deep learning models such as MLP (Multi Layer Perceptron) and TabNet
- Analysis of the effectivenes of the time dependance via. the neural methods like LSTM (Long Short Term Memory)

1.3 Limitations and Weaknesses

- The data imbalance due to high number of no-disease condition compared to sick condition.
- Impossibility associated with the validation of the results with the real time cases.
- Missing value handling in the dataset can cause discontinuity in the time series affecting the temporal pattern.

2 Literature Review

One of the major achievements in the field of Machine Learning is the advancement in the area of disease predictions. These predictions are carried out with respect to medical images, sensory data, or federated data. Among these medical predictions, the disease prediction of animals has gained attention in the recent past. This is due to the fact that there are many regions like Austria, Switzerland, Ireland, and so on, where the majority of investments lie in agriculture and farming, and where medical diagnosis of cattle is a serious problem. This is due to various factors such as (1) availability of medical facilities, (2) remote location of farms, (3) awareness of farmers, and so on. Therefore, it is inevitable for the agriculture industry to have a prediction system that can effectively classify and predict cattle diseases.

One of the successful algorithms developed in the recent past was deployed in the Russian Scientific Agro Engineering Centre, which detects symptoms leading to many diseases such as nervous, infectious diseases, and so on [5]. The studies reported by investigators

faced many issues such as: (1) shortage of data, (2) biased data, (3) generalizability of breeds, and so on. Prior to the above study, Rober et al., 2020 developed a model for predicting mastitis infection patterns in dairy cattle using the SMOTE (Synthetic Minoprity OverSampling Technique) technique to develop synthetic data to adjust for imbalanced data [6]. Even though the bias in the minority cases was handled, the SMOTE technique did not significantly improve the model.

Subsequently, Heidi et al., 2020 approached the problem with an ensemble learning method to predict dairy cattle survival and found that regression ensemble performed better compared to others in the ensemble method [7]. This study was one of its kind where the regression ensemble predicted with the highest precision with AUC (Are Under Curve)ranging from 0.561 to 0.731, followed by the naive ensemble method.

Following this, adapting the Apriori algorithm was another development introduced by Chandana et al., 2020 to help veterinarians diagnose infectious diseases in cattle from electronic records [8]. It was found that the algorithm performed better compared to SVM. However, as usual, imbalance, lower data quantity, and generalizability remained as drawbacks.

Apart from the above studies, a new algorithm was developed for a dataset involving not only cattle features but also environmental features [5]. In addition, prediction of behavioural issues in Austrian dairy cows under subacute ruminal acidosis was attempted by Wagner et al., 2020 [9]. In this study, characteristics of 28 cows were monitored for 9 weeks, and it was found that KNN gave the highest accuracy for detection. However, lesser data content was a drawback.

Following this, with the same dataset, Wagner et al. investigated the reproductive diagnosis of cattle with respect to the daily activity of cows using BOSS (Bag of Symbolic Fourier Approximation Symbols) and FBAT (Fourier Bases Approximation with Thresholding)techniques [10]. BOSS resulted in high speed, and FBAT gave high accuracy. However, generalizability and false alarm presence were the disadvantages of the study.

Furthermore, to avoid limitations in data quantity and quality, a foundation dataset was developed by Lardy et al., 2022 comprising more than 300 cattle [11]. This foundation dataset contains activity patterns and health patterns and was cleaned, curated, and ensured to be viable for machine learning studies. In this case, the absence of features such as environmental data and 17.3 percent missing data were the drawbacks.

Finally, the investigation by Lardy et al., 2023 utilized these 5 datasets together to classify diseases such as lameness, mastitis, acidosis, accident, oestrus, calving, and so on [12]. The classification task aimed to predict the event rather than the disease and found that reproductive events had the highest prediction accuracy accounting for 81% using a Random Forest classifier compared to other classifiers.

In addition to the above disease prediction practices concerning cattle, the methods through which multiple predictions were carried out require an extensive review to understand potential possibilities. Various investigations reported models on multiple predictions via simultaneous multi-output predictions and independent multi-output prediction. The former focuses on predicting multiple outcomes together, and the latter on predicting multiple outcomes independently.

Multilabel chronic disease prediction is one such study reported by Zhang et al., 2019 for the prediction of chronic diseases [13]. GroupNet, GoogLeNet, LSTM (Lomg Short Term Memory), SVM (Support Vector Machine), and Decision Trees were involved in the

comparative analysis for the prediction of 110,300 records with 62 features. Even though GroupNet outperformed other models, the associated computational cost and imbalance of the data were drawbacks.

Joint classification and regression using DM2L (Deep Multi task Multi channel Learning) is another approach adopted by Liu et al., 2019 for predicting Alzheimer's disease [14]. The approach analyzed MRI (Magnetic Resonance Imaging) images and demographic information of patients, including age, gender, and education. Even though the model was efficient, the computational cost was high, and important features such as genetic information, which could improve the accuracy, were absent.

Later in 2021, another multilabel classification for medical purposes was attempted by Liu et al., 2021, for predicting protein subcellular localization [15]. The significant difference in this model compared to other investigations was the incorporation of different feature extraction and dimensionality reduction techniques, which made it more advanced.

Apart from disease prediction studies, multi label prediction is also employed in NLP (Natural Language Processing) for classifying emotions from texts [16]. Multi label KNN (K Nearest Neighbor) is the method used here, which further divides the operations using sample- and label-based MLKNN (Multi Label K Nearest Neighbor) for grammar features in adjacent sentences and the label correlation, respectively. The main drawback of the model is the influence of label correlation on computational cost and lower performance associated with small texts with single labels.

Subsequently, a hybrid multilabel model was introduced by Priyadarshini et al., 2023, with the main aim of compensating for the imbalance contributed by the minority data in datasets [17]. This involved synthetic data incorporation by adaptive synthetic data and ensemble learning. Despite the fact that data imbalance was compensated, the scalability and generalizability of the synthetic data and computational cost added to the cons of this technique.

Apart from the above studies, the multi-classification study by Gopisetti et al., 2023 predicted the presence of various diseases based on patient data using various algorithms [18]. The investigation achieved approximately 99% and 98% accuracy for chronic kidney and diabetes diseases respectively. The former was achieved with the Random Forest, while the latter was achieved using the Decision Tree. Even though the model had high accuracy and potential deployment with Streamlit, information on data imbalance, normalization, feature engineering, and generalizability was lacking.

Furthermore, multi-output and class classification using Gaussian processes was adopted by Chunchao et al., 2023 using a novel approach with linear model coregionalization and Gumbel noise [19]. This advanced method classified each output as a class and incorporated sub-sampling to reduce computational costs. Unlike other models, high scalability is the major highlight of this process. However, feature loss during image downscaling and non-viability in multilabel classification are the cons of this work.

This shortcoming was further investigated by Rahman et al., 2024 in his investigations on breast cancer prediction using machine learning with the help of a multi-output classifier [?].

2.1 Challenges in Previous Studies

- Data imbalance: Majority of the disease shows (0) expressing the healthy condition
- Lack of generalizability: Majority of the studies done on specific breeds which is in

shorted numbers failed to generalize with other breeds.

- Size of the dataset size: the lower size of teh dataset not only reduces the generalizability , but also the scalability and accuracy of predicition.
- Time dependent analysis: The underlying temporal patterns brought out by the cattle with respect tot eh time are least explored.
- Feature correlations: Correlation between teh features are least explored in the previous studies.
- Predictions: The multi output predicitions using deep learning methods are least explored which may capture the underlying complex patterns resulting in accurate analysis.
- Oversampling: the effect of oversampling in the disease prediction by handling the imbalance is least explored.

3 Exploration of data and Methods

4 Proposed Future Analysis

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